AADIT AGRAWAL

Computer Science Engineer · AI/ML · SWEng · HPC · DevOps
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EDUCATION

Manipal Institute of Technology, MAHE — Jul 2023 – May 2027
BTech — Computer and Communication Engineering

EXPERIENCE

Head Student Researcher · *CASIR, MAHE* — May 2025 – Present
- Architected **HPC cluster** (96-core Threadripper 7995WX, 48GB A6000, 256GB RAM) — **1,500+ daily inference req, 99.2% uptime.**
- Leading the setup of a **$200K+ Compute Cluster using Apple Silicon + NVIDIA Blackwell**; optimising for ARM64 + CUDA.
- **Agentic RAG pipeline** with visual indexing + layout-aware OCR — **2,000+ docs, 94% accuracy, 3x faster** than the pre-existing solution.
- **LLM semantic translator** (IEC 61131-3 to IEC 61499) cutting PLC migration by **70%**.
- Real-time CV on **Jetson + DJI drone — 30fps** edge inference via RTMP.

DevOps Engineer, Intern · *Pixa AI, Bengaluru* — Dec 2025
- Provisioned GPU compute (H100/H200/B200) and CI/CD pipelines for TTS/STT model fine-tuning on Phi-4; achieved **99.5% deployment success** across 2 API releases.

Frontend Engineer · *Rossmann New Media 2 Inc.* — Jan 2026 – Present
- Migrated rossmanngroup.com from WordPress to Next.js; improved page load performance and cross-platform responsiveness.
- Implemented technical SEO overhaul targeting data recovery keywords — moved primary rankings from 100+ to top 5.

Founding President · *PRISM* — Jan 2026 – Present
- Research club (AI/ML, SWE, HPC, DevOps, Biotech, InfoSec) — **150+ members**. Academic Advisory Board for AI Curriculum and Adoption.

Management Committee · *MIST* — Oct 2023 – Jun 2025
- CTF infra for **300+ users, 100% uptime**; cut costs by **80%**.
- **12+ workshops** for **150+ participants**; top 10% national CTFs.

RESEARCH

FL + Digital Twin for Route Optimisation · *SCE, MAHE* — Jun 2024 – Jul 2025
- GPS telemetry (**500K+ datapoints**); dashcam CV via OpenCV + YOLO — **12,000+ labelled samples.**
- Built FL pipeline for privacy-preserving route optimization across distributed edge nodes.

PROJECTS

Homelab Infrastructure · *Self-hosted cloud* — 2022 – Present
- **70+ services** on Proxmox/Docker — **98% uptime, 4 yrs**, 8TB MergerFS, Blackwell-GPU, 3-2-1 backups.
- Deployed custom object-detection models on security camera feeds achieving **30fps** real-time inference; offloaded **95%** of compute from CPU to GPU including 4K media transcoding.
- Mitigated **live DDoS (50K+ req/min)**; **95%** attack surface reduction.

The TIK Menu API · GitHub — 2025 – Present
- **5,000+ DAU, 150K+ monthly req**, campus adoption in under 7 days. REST API + Cloudflare R2.

Coolstuff.work Forgejo — 2025 – Present
- Built **OR-Free**, a real-time AI model availability and pricing tracker for OpenRouter, and **MIT Courses Search**, a course discovery tool with interactive graph visualization (**500+ nodes**). Next.js, Radix UI.
- Additional tools: transcript-cli (video transcription), Campus Directory (crowdsourced contacts), Mass Contact Maker (bulk VCF generation). See git.aadit.cc.

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Go, Zig, Rust, Bash, SQL
Infrastructure & DevOps: Docker, Kubernetes, Proxmox, Linux, Git, GitHub Actions, Terraform, Ansible, Nginx, Caddy, Tailscale
Cloud & Platforms: AWS, GCP, Azure, Hetzner, Cloudflare Workers, Vercel
AI/ML & HPC: PyTorch, JAX, Transformers, CUDA, SGLang, vLLM, llama.cpp, TensorRT-LLM, OpenCV, YOLO, Slurm, NCCL
Web: Next.js, React, Svelte, Astro, Tailwind, Node.js, Bun, FastAPI, tRPC
Databases: PostgreSQL, MongoDB, SQLite, Redis, pgvector, ChromaDB
AI Tooling: Pi (Custom Implementation), Codex, Claude Code, Cursor, MCP, Skills